



BANOL[®]

Fungicide

For control of Pythium and Phytophthora

ACTIVE INGREDIENT:

Propamocarb hydrochloride* 66.5%

OTHER INGREDIENTS:..... 33.5%

TOTAL: 100.0%

*Propyl [3-(dimethylamino)propyl]
carbamate monohydrochloride

Contains 6.0 lbs. active ingredient per gallon
Shake Well Before Use

EPA Reg. No. 101563-21

**STOP - READ THE LABEL
BEFORE USE**

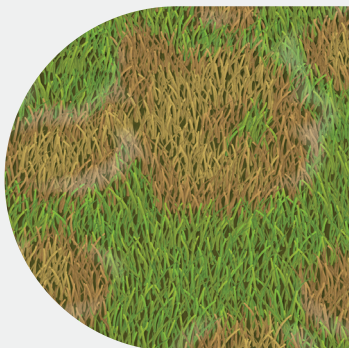
**KEEP OUT OF REACH OF
CHILDREN**

CAUTION

**For MEDICAL and TRANSPORTATION
Emergencies ONLY Call 24 Hours A Day
1-800-424-9300**

**For PRODUCT USE Information Call
1-800-331-2867**

See Back Panel for First Aid Instructions and
Booklet for Complete Precautionary
Statements and Directions for Use.



FIRST AID

If swallowed:	• Call a poison control center or doctor immediately for treatment advice. • Have person sip a glass of water if able to swallow. • Do not induce vomiting unless told to by a poison control center or doctor. • Do not give anything by mouth to an unconscious person.
If in eyes:	• Hold eyes open and rinse slowly and gently with water for 15 - 20 minutes. • Remove contact lenses, if present, after the first 5 minutes, then continue rinsing. • Call a poison control center or doctor for treatment advice.
If on skin or clothing:	• Take off contaminated clothing. • Rinse skin immediately with plenty of water for 15 - 20 minutes. • Call a poison control center or doctor for treatment advice.

NOTE TO PHYSICIAN: Do not administer atropine unless cholinesterase inhibition is measured or patient shows distinct symptoms thereof. Oxime therapy is contraindicated. Otherwise, administration of gastric lavage or induction of vomiting followed by application of activated charcoal is the recommended treatment.

IN CASE OF MEDICAL EMERGENCIES OR HEALTH AND SAFETY INQUIRIES, OR IN CASE OF FIRE, LEAKING, OR DAMAGED CONTAINERS, INFORMATION MAY BE OBTAINED BY CALLING 1-800-424-9300. HAVE THE PRODUCT CONTAINER OR LABEL WITH YOU WHEN CALLING A POISON CONTROL CENTER OR DOCTOR, OR GOING FOR TREATMENT.

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

CAUTION

Harmful if swallowed, or absorbed through the skin. Causes moderate eye irritation. Avoid contact with skin, eyes, or clothing. Prolonged or frequently repeated skin contact may cause allergic reactions in some individuals.

PERSONAL PROTECTIVE EQUIPMENT

Applicators and other handlers must wear:

- Long-sleeved shirt and long pants
- Chemical-resistant gloves made of any waterproof material such as polyethylene or polyvinyl chloride
- Shoes and socks
- Chemical resistant apron when making dip applications

Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables exist, use detergent and hot water. Keep and wash PPE separately from other laundry.

ENGINEERING CONTROLS STATEMENT

When handlers use closed systems or enclosed cabs in a manner that meets the requirements listed in the Worker Protection Standard (WPS) for agricultural pesticides [40 CFR 170.240 (d) (4-6)], the handler PPE requirements may be reduced or modified as specified in the WPS.

USER SAFETY RECOMMENDATIONS

Users should:

- Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet.
- Users should remove clothing if pesticide gets inside. Then wash thoroughly and put on clean clothing.
- Users should remove protective clothing and equipment immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing.

ENVIRONMENTAL HAZARDS

Do not apply directly to water, or to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwaters. Do not apply when weather conditions favor drift from treated areas.

DIRECTIONS FOR USE

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

Do not apply this product in a way that will contact workers or other persons either directly or through drift. Only protected handlers may be in the area during application. For any requirements specific to your State or Tribe, consult the agency responsible for pesticide regulation.

AGRICULTURAL USE REQUIREMENTS

Use this product only in accordance with its labeling and with the Worker Protection Standard 40 CFR 170. This Standard contains requirements for the protection of agricultural workers on farms, forests, nurseries, and greenhouses, and handlers of agricultural pesticides. It contains requirements for training, decontamination, notification, and emergency assistance. It also contains specific personal protective equipment and restricted entry interval. The requirements in this box only apply to uses of this product that are covered by the Worker Protection Standard.

Do not enter or allow worker entry into treated areas during the restricted entry interval (REI) of 12 hours. **Exception:** If the product is soil injected, soil incorporated, or applied by drenching, the Worker Protection Standard under certain circumstances, allows workers to enter the treated area without restrictions if there will be no contact with anything that has been treated.

PPE required for early entry to treated areas that is permitted under the Worker Protection Standard and that involves contact with anything that has been treated, such as plants, soil, or water, is:

- Coveralls
- Shoes and socks
- Chemical-resistant gloves made of any waterproof material

NON-AGRICULTURAL USE REQUIREMENTS

The requirements in this box apply to uses of this product that are not within the scope of the Worker Protection Standard for agricultural pesticides (40 CFR Part 170). The WPS applies when this product is used to produce agricultural plants on farms, forests, nurseries, or greenhouses.

Do not enter or allow others to enter treated areas until sprays have dried.

PRODUCT INFORMATION

BANOL® FUNGICIDE is a completely water miscible fungicide concentrate. BANOL FUNGICIDE is intended for use on turfgrass or ornamental species which are subject to Pythium Blight (Cottony Blight, Grease Spot) or *Phytophthora*. It may be used after germination to control Pythium damping-off in turf, or on established turf to prevent or control Pythium Blight. BANOL FUNGICIDE does not afford broad spectrum disease control for *Rhizoctonia* sp. (brown patch), *Clarireedia* sp. (syn. *Sclerotinia* sp.) (dollar spot), or *Helminthosporium* sp. (leafspot). Use a broad spectrum fungicide if these diseases coexist with Pythium Blight.

BANOL FUNGICIDE is also labeled for prevention of root rot and damping-off caused by *Pythium* sp. and *Phytophthora* sp. at all stages of ornamental propagation.

COMPATIBILITY AND MIXING:

BANOL FUNGICIDE is compatible with many pesticides and liquid fertilizers. A compatibility test must be conducted with any potential tank-mix partner with BANOL FUNGICIDE. Using a clear container, conduct the test as described below:

1. Fill the container three-quarters full with water.
2. Add the products to be tank-mixed in the following order: (a) wettable powders (b) dry flowable, (c) aqueous suspensions, (d) BANOL FUNGICIDE*, (e) liquids, (f) solutions and emulsifiable liquid concentrates, (g) micronutrients and liquid fertilizers.
3. Shake or stir after each addition to mix thoroughly.
4. After adding all ingredients, let the mixture stand for 15 minutes and look for separation, large flakes, precipitates, gels, and heavy oily film or other signs of incompatibility.
5. If the compatibility test shows signs of incompatibility, do not tank-mix the product tested with BANOL FUNGICIDE.

*Note that a labeled application rate for BANOL FUNGICIDE must be used in any compatibility test. Dilute from a concentrated suspension, or sample from a properly diluted spray solution.

Follow use precautions and restrictions on this label and for all tank-mix partners, if mixing BANOL FUNGICIDE with other pesticides.

BANOL FUNGICIDE requires no agitation after initial mixing. Stock solutions of BANOL FUNGICIDE must be used within one day of mixing. It is the pesticide user's responsibility to ensure that all products are registered for the intended use. Read and follow the applicable restrictions and limitations and directions for use on all product labels involved in tank mixing. Users must follow the most restrictive directions for use and precautionary statements of each product in the tank mixture.

USE RESTRICTIONS

- Do not graze or feed clippings from treated areas to livestock.
- Do not apply this product through any type of irrigation system.
- Do not apply this product by aerial application.

SPRAY DRIFT ADVISORIES

THE APPLICATOR IS RESPONSIBLE FOR AVOIDING OFF-SITE SPRAY DRIFT. BE AWARE OF NEARBY NON-TARGET SITES AND ENVIRONMENTAL CONDITIONS.

IMPORTANCE OF DROPLET SIZE

An effective way to reduce spray drift is to apply large droplets. Use the largest droplet size that provides target pest control. While applying larger droplets will reduce spray drift, the potential drift will be greater if applications are made improperly or under unfavorable environmental conditions.

Controlling Droplet Size

Volume - Increasing the spray volume so that larger droplets are produced will reduce spray drift. Use the highest practical spray volume for the application. If a greater spray volume is needed, consider using a nozzle with a higher flow rate.

Pressure - Use the lowest spray pressure recommended for the nozzle to produce the target spray volume and droplet size.

Spray Nozzle - Use a spray nozzle that is designed for the intended application. Consider using nozzles designed to reduce drift.

RELEASE HEIGHT

For ground equipment, the boom should remain level with the target area and have minimal bounce.

SHIELDED SPRAYERS

Shielding the boom or individual nozzles can reduce spray drift. Consider using shielded sprayers. Verify that the shields are not interfering with the uniform deposition of the spray on the target area.

TEMPERATURE INVERSIONS

Drift potential is high during a temperature inversion. Temperature inversions are characterized by increasing temperature with altitude and are common on nights with limited cloud cover and light to no wind. The presence of an inversion can be indicated by ground fog or by the movement of smoke from a ground source smoke generator. Smoke that layers and moves laterally in a concentrated cloud (under low wind conditions) indicates an inversion, while smoke that moves upwards and rapidly dissipates indicates good vertical air mixing. Avoid applications during temperature inversions.

WIND

Drift potential generally increases with wind speed. AVOID APPLICATIONS DURING GUSTY WIND CONDITIONS. Applicators need to be familiar with local wind patterns and terrain that could affect spray drift.

Boomless Ground Applications

Setting nozzles at the lowest effective height will help to reduce the potential for spray drift.

Hand-held Technology Applications

Take precautions to minimize spray drift.

RESISTANCE MANAGEMENT

For resistance management, BANOL FUNGICIDE contains a group 28 fungicide. Any fungal population may contain individuals naturally resistant to BANOL FUNGICIDE and other group 28 fungicides. A gradual or total loss of pest control may occur over time if these fungicides are used repeatedly in the same fields. Appropriate resistance management strategies should be followed.

To delay fungicide resistance, take one or more of the following steps:

- Use tank mixtures with fungicides from a different group that are equally effective on the target pest when such use is permitted. Use at least the minimum application rate as labeled by the manufacturer.
- Adopt an integrated disease management program for fungicide use that includes scouting, uses historical information related to pesticide use, and crop rotation, and which considers host plant resistance, impact of environmental conditions on disease development, disease thresholds, as well as cultural, biological and other chemical control practices.
- Where possible, make use of predictive disease models to effectively time fungicide applications. Note that using predictive models alone

is not sufficient to manage resistance.

- Monitor treated fungal populations for resistance development.
- For further information or to report suspected resistance contact Environmental Science U.S., LLC at 1-800-331-2867. You can also contact your pesticide distributor or university extension specialist to report resistance.

TURFGRASS USE INSTRUCTIONS

BANOL FUNGICIDE is labeled for prevention of root rot and damping-off caused by *Pythium* sp. and *Phytophthora* sp. at all stages of turfgrass development.

TURFGRASS*	AREA	FLUID OUNCES BANOL FUNGICIDE	REMARKS
For Control of: Pythium Blight, Pythium Damping Off, and Root Rot on Established Turf and Overseeded Areas	1,000 sq ft	Preventative Treatment* 1-1/3 - 2 fl oz in 2 - 5 gallons of water Curative Treatment* 3 - 4 fl oz in 2 - 5 gallons of water	ESTABLISHED TURF: Apply as a preventative treatment during periods of high temperature and humidity favorable to the development of Pythium Blight. OVERSEEDED AREAS: Apply after germination to control Pythium Damping-Off. RETREATMENT: Re-treat at 7 - 21 day intervals if conditions remain favorable for disease development. Use lower rate for the shorter interval and the higher rate for the longer interval within the application rate range. Under severe disease pressure conditions, use the highest labeled rate and shortest interval.
* 1 1/3, 2, 3, and 4 fl oz of product per 1000 sq ft is equivalent to 2.72, 4.08, 6.13, and 8.17 lbs ai per acre, respectively			
Turfgrass Use Restrictions • Do not apply more than a total of 12.25 fl oz (0.57 lb ai) of BANOL FUNGICIDE per 1,000 sq ft of turfgrass per year, equivalent to 25 lb ai/A/year. • Not for use on Sod Farms in Arizona.			

ORNAMENTAL USE INSTRUCTIONS

Apply to commercially grown ornamentals in greenhouses and container grown ornamentals in nurseries by soil-directed drench only. BANOL FUNGICIDE does not cause phytotoxic symptoms on the listed plants when used as directed. Trials on other ornamental species must be conducted before extensive use. Do not apply a soil drench to the foliage.

Use chemical resistant gloves when planting cuttings, seedlings, or seeds into the potting media that is still wet from a BANOL FUNGICIDE drench application.

ORNAMENTALS	FLUID OUNCES BANOL FUNGICIDE	REMARKS
For Control of: <i>Pythium</i> and <i>Phytophthora</i> on Ornamental Plants: African Violet, Ageratum, Alyssum, Aster, Azalea, Browallia, Carnation, Celosia, Chrysanthemum, Coleus, Dahlia, Dusty Miller, Easter Lily, Ferns, Geranium, Gloxinia, Impatiens, Lobelia, Marigold, Pansy, Petunia, Poinsettia, Portulaca, Salvia, Snapdragon, Verbena, Violet, and Vinca.	30 fl oz per 100 gallons of water 30 fl oz per 100 gallons of water 20 fl oz per 100 gallons of water 20 fl oz per 100 gallons of water	Seeding: Apply to the seeding area using 3 quarts of solution / 10 sq ft. Treat before sowing of seeds requiring light and after covering of seeds requiring dark for germination. Transplanting (Seedlings): Apply to the transplanting area using 3 quarts of solution / 10 sq ft before transplanting. Transplanting (Cuttings): Drench the cutting bed using 2 gallons of solution / 10 sq ft before setting. Potting: Dip bare root plants in the solution before potting. After potting drench each pot using 3.5 fl oz of solution per 4 inch pot.
Woody Ornamentals	25 fl oz per 100 gallons of water	For control of <i>Phytophthora</i> sp. on woody ornamentals, drench each container at the rate of 5 quarts of solution/10 sq ft of surface.
Ornamental Use Restrictions • Do not use for field grown ornamentals. • Do not apply more than 63 fl oz of product (2.95 lbs ai) per 1000 square feet per crop production cycle for greenhouses. • Do not make more than 2 applications per crop production cycle. If a second application is needed, the minimum retreatment interval is 7 days.		

STORAGE AND DISPOSAL

Do not contaminate water, food or feed by storage or disposal.

PESTICIDE STORAGE

Store in original container and keep tightly closed when not in use. Store in a cool, dry place. Avoid cross-contamination with other pesticides.

PESTICIDE DISPOSAL

Pesticides wastes are toxic. Improper disposal of excess pesticide, spray mixture, or rinsate is a violation of Federal Law. If these wastes cannot be used according to label instructions, contact your State Pesticide or Environmental Control Agency or Hazardous Waste representative at the nearest EPA regional office for guidance in proper disposal methods.

CONTAINER HANDLING

Non-Refillable Containers

Rigid, Non-refillable containers (equal to or less than 5 gallons)

Non-refillable container. Do not reuse or refill this container. Triple rinse or pressure rinse container (or equivalent) promptly after emptying. Triple rinse as follows: Empty the remaining contents into application equipment or a mix tank and drain for 10 seconds after the flow begins to drip. Fill the container 1/4 full with water and recap. Shake for 10 seconds. Pour rinsate into application equipment or a mix tank or store rinsate for later use or disposal. Drain for 10 seconds after the flow begins to drip. Repeat this procedure two more times. Pressure rinse as follows: Empty the remaining contents into application equipment or a mix tank and continue to drain for 10 seconds after the flow begins to drip. Hold container upside down over application equipment or mix tank or collect rinsate for later use or disposal. Insert pressure rinsing nozzle in the side of the container, and rinse at about 40 PSI for at least 30 seconds. Drain for 10 seconds after the flow begins to drip.

Once container is rinsed, offer for recycling if available or puncture and dispose of in a sanitary landfill or by incineration.

Rigid Non-refillable Containers that are too large to shake (i.e., with capacities greater than 5 gallons or 50 lbs)

Non-refillable container. Do not reuse or refill this container. Refer to Bottom Discharge IBC or Top Discharge IBC, Drums, Kegs information as follows.

Bottom Discharge IBC (e.g. – Schuetz Caged IBC or Snyder Square Stackable)

Pressure rinsing the container before final disposal is the responsibility of the person disposing of the container. To pressure rinse the container before final disposal, empty the remaining contents from the IBC into application equipment or mix tank. Raise the bottom of the IBC by 1.5 inches on the side which is opposite of the bottom discharge valve to promote more complete product removal. Completely remove the top lid of the IBC. Use water pressurized to at least 40 PSI to rinse all interior portions. Continuously pump or drain rinsate into application equipment or rinsate collection system while pressure rinsing. Continue pressure rinsing for 2 minutes or until rinsate becomes clear. Replace the lid and close bottom valve.

Top Discharge IBC, Drums, Kegs (e.g. – Snyder 120 Next Gen, Bonar B120, Drums, and Kegs)

Triple rinsing the container before final disposal is the responsibility of the person disposing of the container. To triple rinse the container before final disposal, empty the remaining contents from this container into application equipment or mix tank. Fill the container at least 10 percent full with water. Agitate vigorously or recirculate water with the pump for 2 minutes. Rinse all interior surfaces. Pour or pump rinsate into application equipment or rinsate collection system. Repeat this procedure two more times.

Once container is rinsed, offer for recycling if available or puncture and dispose of in a sanitary landfill or by incineration.

Non-Refillable Fiber Drums with Liners

Non-refillable container. Do not reuse or refill this container. Completely empty liner by shaking and tapping sides and bottom to loosen clinging particles. Empty residue into application equipment, then offer for recycling if available or dispose of in a sanitary landfill or by incineration. If drum is contaminated and cannot be reused, dispose of it in the manner required for its liner.

Non-Rigid, Non-refillable Containers

Nonrefillable container. Do not reuse or refill this container. Completely empty container into application equipment. Then offer for recycling if available or dispose of in a sanitary landfill or by other procedures approved by state and local authorities."

Refillable Containers

Refillable container. Refer to Bottom Discharge IBC or Top Discharge IBC, Drums, Kegs information as follows. Refill this container with pesticide only. Do not reuse this container for any other purpose. Contact your Ag retailer or Environmental Science U.S., LLC for container return, disposal and recycling information.

Bottom Discharge IBC (e.g. – Schuetz Caged IBC or Snyder Square Stackable)

Pressure rinsing the container before final disposal is the responsibility of the person disposing of the container. Cleaning before refilling is the responsibility of the refiller. To pressure rinse the container before final disposal, empty the remaining contents from the IBC into application equipment or mix tank. Raise the bottom of the IBC by 1.5 inches on the side which is opposite of the bottom discharge valve to promote more complete product removal. Completely remove the top lid of the IBC. Use water pressurized to at least 40 PSI to rinse all interior portions. Continuously pump or drain rinsate into application equipment or rinsate collection system while pressure rinsing. Continue pressure rinsing for 2 minutes or until rinsate becomes clear. Replace the lid and close bottom valve.

Top Discharge IBC, Drums, Kegs (e.g. – Snyder 120 Next Gen, Bonar B120, Drums, Kegs)

Triple rinsing the container before final disposal is the responsibility of the person disposing of the container. Cleaning before refilling is the responsibility of the refiller. To triple rinse the containers before final disposal, empty the remaining contents from this container into application equipment or mix tank. Fill the container at least 10 percent full with water. Agitate vigorously or recirculate water with the pump for 2 minutes. Rinse all interior surfaces. Pour or pump rinsate into application equipment or rinsate collection system. Repeat this procedure two more times.

Once container is rinsed, offer for recycling if available or puncture and dispose of in a sanitary landfill or by incineration.

End users are authorized to remove tamper-evident cables as required to remove the product from the container unless the container is equipped with one-way valves and refilling or returning is planned. If this is the case, end-users are not authorized to remove tamper-evident cables, remove one-way valves, or clean container.

IMPORTANT: READ BEFORE USE

Read the entire Directions for Use, Conditions, Disclaimer of Warranties and Limitations of Liability before using this product. If terms are not acceptable, return the unopened product container at once.

By using this product, user or buyer accepts the following Conditions, Disclaimer of Warranties and Limitations of Liability.

CONDITIONS: The directions for use of this product are believed to be adequate and must be followed carefully. However, it is impossible to eliminate all risks associated with the use of this product. Crop injury, ineffectiveness or other unintended consequences may result because of such factors as weather conditions, presence of other materials, or the manner of use or application, all of which are beyond the control of Environmental Science U.S., LLC. All such risks shall be assumed by the user or buyer.

DISCLAIMER OF WARRANTIES: TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, ENVIRONMENTAL SCIENCE U.S., LLC MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OF MERCHANTABILITY OR OF FITNESS FOR A PARTICULAR PURPOSE OR OTHERWISE, THAT EXTEND BEYOND THE STATEMENTS MADE ON THIS LABEL. No agent of Environmental Science U.S., LLC is authorized to make any warranties beyond those contained herein or to modify the warranties contained herein. TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, ENVIRONMENTAL SCIENCE U.S., LLC DISCLAIMS ANY LIABILITY WHATSOEVER FOR SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT.

LIMITATIONS OF LIABILITY: TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, THE EXCLUSIVE REMEDY OF THE USER OR BUYER FOR ANY AND ALL LOSSES, INJURIES OR DAMAGES RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT, WHETHER IN CONTRACT, WARRANTY, TORT, NEGLIGENCE, STRICT LIABILITY OR OTHERWISE, SHALL NOT EXCEED THE PURCHASE PRICE PAID, OR AT ENVIRONMENTAL SCIENCE U.S., LLC'S ELECTION, THE REPLACEMENT OF PRODUCT.

Envu and the Envu logo are trademarks and Banol® is a registered trademark owned by Environmental Science U.S., LLC or one of its affiliates.

PRODUCED FOR
Environmental Science U.S., LLC
5000 CentreGreen Way, Suite 400
Cary, NC 27513

envu™

Specimen Label